



Operator #: 2487

Operation Name: Randy & Whilden Hughes

Date: 05/21/2025

As you describe your natural resource management practices, please note that some of your answers to questions about natural resource management may be relevant to other sections of this OSP as they relate to crop rotation, nutrient and manure management. The following questions are intended to address general, whole-farm goals and approaches to organic natural resource management.

1. Please describe how your farming or ranching practices – crop and/or livestock production practices – maintain or improve natural resources, foster cycling of nutrients, promote ecological balance, and conserve biodiversity. Please list the specific strategies or practices you use that contribute to each general goal listed below, as applicable.

A) Conserve soil; Improve soil condition; Prevent soil erosion.

Manage waterways, manage tiling, plant on the contour, extensive cover crops, implement tillage only when necessary, try very hard to have “roots in the ground year round”

We maintain and improve natural resources by always working to prevent runoff and leaching. We do timed applications of fertility and variable rate. Our applications are always based off of recent soil samples and removal rates. We rarely remove straw, unless to reduce P&K levels in soil. We plant on the contour when necessary

We foster nutrient cycling by applications of liquid biologicals and disking down cover crops and biomass.

We are conserving biodiversity and ecological balance by applying for CSP programs to extend managed fence lines to 30 ft on kopp farm and swag farm

B) Prevent water contamination by plant nutrients; Protect water quality.

Buffer strips along streams. Enrolled in CSP and EQIP programs through NRCS. we have begun to sidedress compost pellets to space out our nitrogen applications. extensive cover cropping.



C) Conserve water

variable rate irrigation, rainfall tracking via fieldview, drop nozzles, Low PSI pumps, solar powered moisture probes on farm. Extensive cover cropping improves water retention

D) Conserve biodiversity (soil organisms, pollinators, natural enemies of pests, predators, native habitat, vegetation and wildlife).

Manage fence lines, maintain wooded non-production land, enroll land in Driftwatch program. Keep diverse crop rotations. Managed marshes and woodlands. We have also started managing bee hives to increase pollinator population, recently Partnering with University of Massachusetts to study effects of sunflower pollen on pollinator health.

E) Other practices that maintain or improve natural resources (soil, water, wetlands, woodlands and wildlife), foster cycling of nutrients, promote ecological balance, and conserve biodiversity.

☐ **Not Applicable**

20 kw solar panels produce enough energy to offset the power requirements of our shop.
Used oil burner recycles spent oil into heat, smart LED lighting, and skylights.